

TGFB1 promoter-reporter architectures

5' -> 3' display, genomic 41358961 -> 41287203 | strand - | assembly GRCh38 | annotation Ensembl live GRCh38 | upstream 5000 bp | downstream 1000 bp

Canonical reporter comparison: 1 TSS class(es), 3 transcript(s), 9 explicit architecture row(s); normalized evidence remains observational/predictive context.

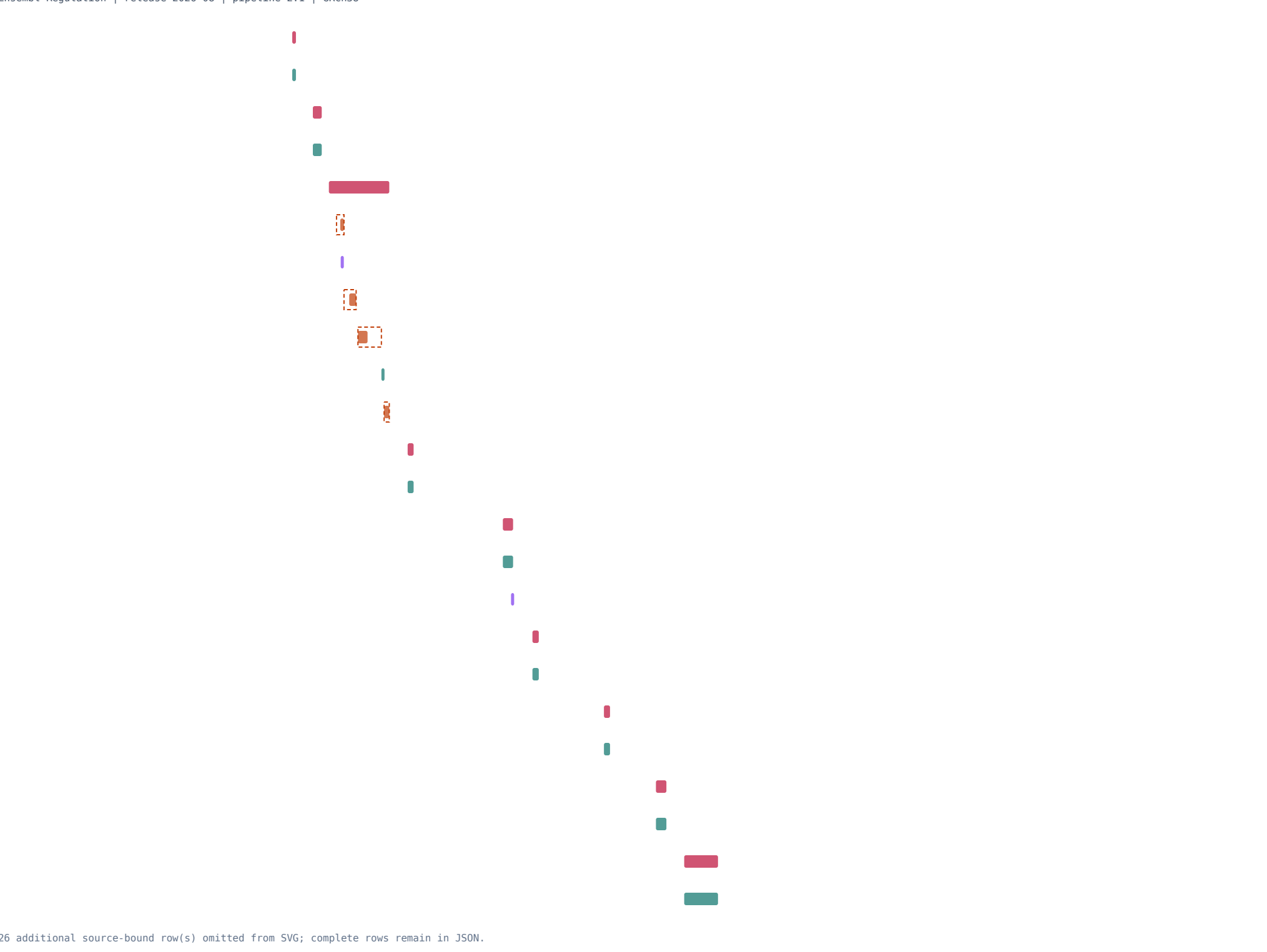


Proposed reporter architectures



Ensembl-annotated regulatory regions

Ensembl Regulation | release 2026-08 | pipeline 2.1 | GRCh38



26 additional source-bound rows omitted from SVG; complete rows remain in JSOM.

Projected occupancy grouped by declared biological context

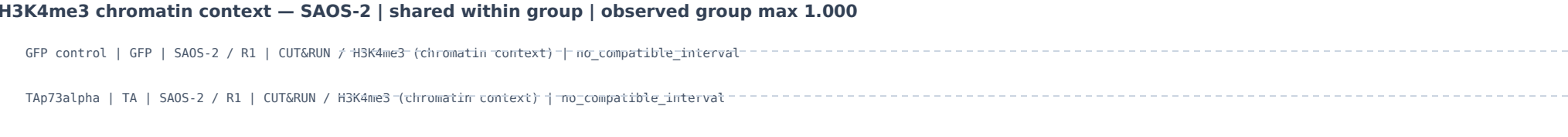
TP73 occupancy — SAOS-2 | shared within group | observed group max 1.000



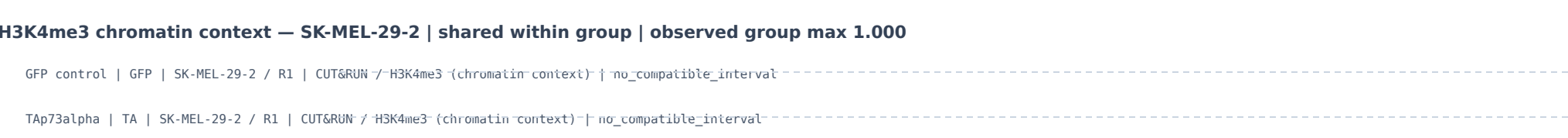
TP73 occupancy — SK-MEL-29-2 | shared within group | observed group max 1.000



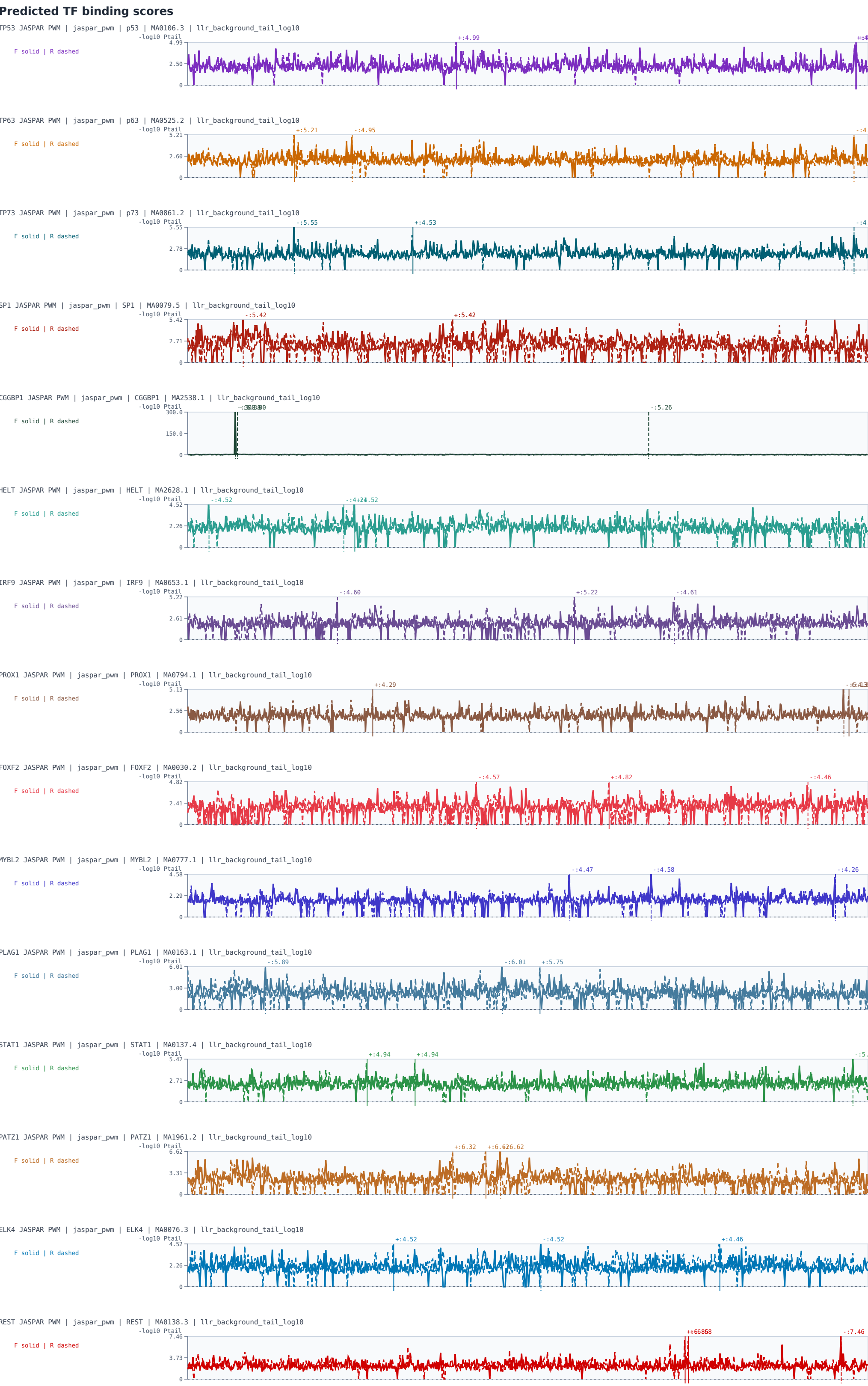
H3K4me3 chromatin context — SAOS-2 | shared within group | observed group max 1.000



H3K4me3 chromatin context — SK-MEL-29-2 | shared within group | observed group max 1.000



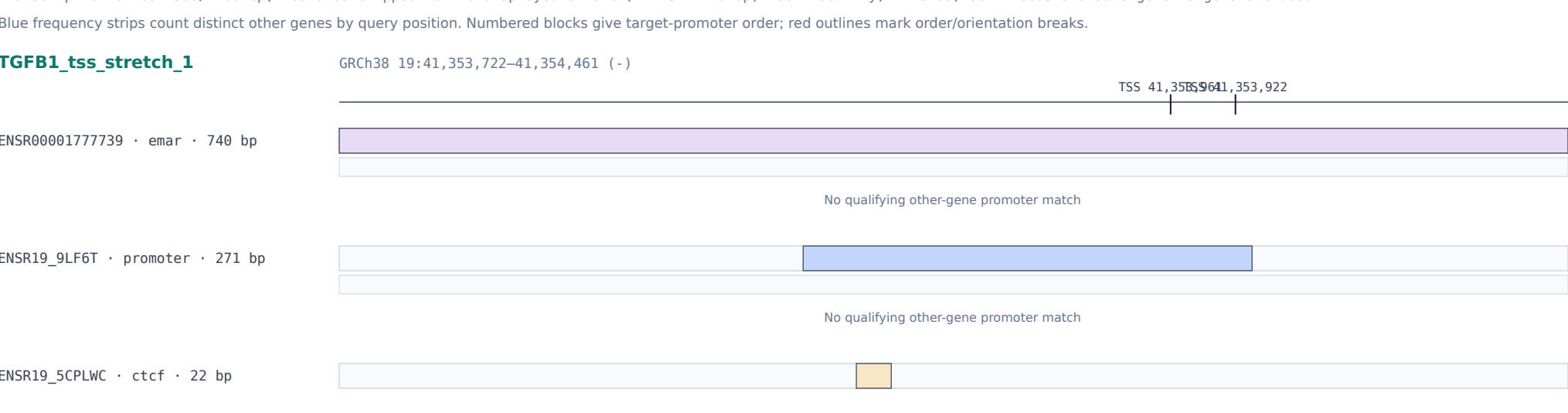
Predicted TF binding scores



TSS-local Ensembl-feature promoterome similarity

Transcript-oriented ~500/4200 bp; features clipped to the displayed stretch; BLASTN ≥40 bp, ≥80% identity, Exle-05; self-locus and same-gene targets excluded.

Blue frequency strips count distinct other genes by query position. Numbered blocks give target-promoter order; red outlines mark order/orientation breaks.



Reporter interpretation boundaries

Transcript annotation alone does not establish which TSS is used in the assayed cells.

CUT&RUN overlap is occupancy/enrichment evidence, not proof of TSS usage, promoter activity, direct regulation, biochemical affinity, or luciferase output.

Reporter geometry does not verify the identity of a physical laboratory construct.

Warnings / interpretation checks

warning: Occupancy lane 'SAOS-2 H3K4me3 Dnp73beta' is retained as no_compatible_interval; no compatible interval is rendered.
warning: Occupancy lane 'SAOS-2 H3K4me3 GFP' is retained as no_compatible_interval; no compatible interval is rendered.
warning: Occupancy lane 'SAOS-2 H3K4me3 Tap73alpha' is retained as no_compatible_interval; no compatible interval is rendered.
warning: Occupancy lane 'SAOS-2 TP73 Dnp73beta' is retained as no_compatible_interval; no compatible interval is rendered.
warning: Occupancy lane 'SAOS-2 TP73 GFP' is retained as no_compatible_interval; no compatible interval is rendered.
warning: Occupancy lane 'SAOS-2 TP73 Tap73alpha' is retained as no_compatible_interval; no compatible interval is rendered.
warning: Occupancy lane 'SK-MEL-29-2 H3K4me3 Dnp73beta' is retained as no_compatible_interval; no compatible interval is rendered.
warning: Occupancy lane 'SK-MEL-29-2 H3K4me3 GFP' is retained as no_compatible_interval; no compatible interval is rendered.

Evidence provenance

ensembl_regulation_annotation: ensembl_regulation_2026_08_grch38
ensembl_regulation_annotation_intervals: ensembl_regulation_2026_08_grch38
isoform_panel: TGFB1_panel
occupancy_layout: tp73_saos2,tp73_sknel29_2,h3k4me3_saos2,h3k4me3_sknel29_2
predicted_tf_score_jaspar_pwm: local_jaspar_registry
requested_occupancy_track: E-MTAB-15789:h3k4me3:saos2:DN:R1
requested_occupancy_track: E-MTAB-15789:h3k4me3:saos2:GFP:R1
requested_occupancy_track: E-MTAB-15789:h3k4me3:sknel29-2:DN:R1
requested_occupancy_track: E-MTAB-15789:h3k4me3:sknel29-2:GFP:R1